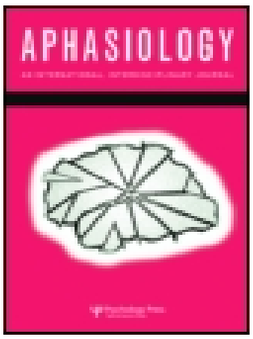

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Revised administration and scoring procedures for the Boston Naming Test and norms for non-brain-damaged adults

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Abstract

Standard administration, response coding and scoring procedures were developed for the 60-item Boston Naming Test (BNT). Standard prompts to be given following certain types of incorrect responses were also developed. The BNT was administered to 60 non-brain-damaged adults using these standard procedures. Our results show that examiners can reliably carry out the revised test procedures reported in this paper. Our results also show that two response categories (Related Name and Don't Know) accounted for nearly 80% of incorrect responses. Several BNT test pictures frequently elicited a consistent name other than the one specified in the test. Prompting for an additional response following Related Name and Multiple Attempts responses frequently elicited the correct BNT name. Prompts following other types of error responses usually did not elicit the correct name. Neither the age nor education of our nonbrain-damaged subjects was strongly related to their total BNT score.

Introduction

The Boston Naming Test (BNT; Kaplan, Goodglass and Weintraub 1983) is a 60-item confrontation naming test which is widely used to assess the word retrieval performance of adults with brain damage. Kaplan *et al.* (1983) reported normative information for subjects up to the age of 59 years old. These subjects had actually been tested with an earlier 85-item version of the BNT. Their scores were then reanalysed looking only at the 60 items contained in the 1983 version, and these interpolated data were published with the 1983 test. Kaplan *et al.* provide no information about item-by-item error rates or about the kinds of incorrect responses that nonbrain-damaged subjects can be expected to give on the test. Van Gorp, Satz, Kiersch and Henry (1986) subsequently published normative data on the (60-item) Boston Naming Test performance of 78 nonbrain-damaged adults between the ages of 59 and 95 years.

The Boston Naming Test is, as far a we know, the only commercially available test designed exlusively to assess the confrontation naming of brain-damaged adults.

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However, test administration and response scoring instructions are not specific enough or comprehensive enough to ensure that users of the test will administer the test and score responses to test items in the same way. No interjudge or intrajudge reliability data have been reported for the Boston Naming Test. Consequently, users cannot assume that different test scores obtained by the same examiner on two occasions, or by different examiners testing the same patient, truly represent changes in the patients's naming performance and not differences in how the test was administered. Some pictures from the Boston Naming Test often elicit responses that seem reasonable names for the pictures but do not appear to be correct according to scoring instructions for the test. If some test items frequently elicit synonyms for test items from non-brain-damaged adults, then it seems reasonable to consider those synonyms acceptable responses to the test items.

The Boston Naming Test was designed so that the familiarity of the pictured items decreases as one progresses through the test. Consequently, one would expect that non-brain-damaged adults would make few errors on early test items, but that error rates would increase toward the end of the test. However, a few stimulus pictures from the test appear to be ambiguous or visually confusing. Even non-brain-damaged adults might have difficulty naming them, not because they are unfamiliar with the name of the pictures, but because they are confused about what the pictures represent. Several of these apparently ambiguous or confusing pictures occur early in the test. (e.g., item No. 4, 'house', which often elicits names such as 'apartment building' or 'schoolhouse').

The Boston Naming Test permits two kinds of prompts from examiners. If the patient's response to a test item suggests that he or she has misperceived the picture, the examiner provides a *stimulus cue* (a phrase providing additional information about the referent). If the patient cannot provide the correct name for a test item within 20 seconds, the examiner can give a *phonemic cue* (the initial phoneme(s) of the item). Credit is given if the patient correctly names a test item within 20 seconds, either spontaneously or following a stimulus cue, but no credit is given if the correct name is produced after a phonemic cue. It seems likely that one might learn more about patients' word retrieval abilities if they are prompted to produce another response following some incorrect responses.

In this report we describe a number of modifications of standard Boston Naming Test procedures. We felt that these changes would improve the interexaminer and intraexaminer reliability of response scoring, and provide additional information about patients' word retrieval abilities The modifications include:

(1) Standard administration procedures.
(2) Standard response coding and scoring procedures.
(3) Standard prompts to be given following certain types of incorrect responses.

We also report norms and reliability information for the Boston Naming Test for 60 healthy non-brain-damaged adults. We analysed the performance of the non-brain-damaged adults to determine the following:

(1) Item-by-item error rates for the 60 test items.
(2) The types of incorrect responses that non-brain-damaged subjects give to Boston Naming Test items.
(3) How subjects respond when prompts are given following incorrect responses.

Part I: Evaluation of administration and coding procedures

Method

(1) Testing non-brain-damaged subjects

Subjects

Subjects were 60 non-brain-damaged adults of 40–78 years of age ($M = 56{\cdot}1$, $SD = 9{\cdot}17$). The length of the subjects' education ranged from 9 to 20 years ($M = 13{\cdot}95$, $SD = 2{\cdot}30$). They were recruited from veterans' organizations and their auxiliaries in the Minneapolis–St. Paul metropolitan area. None were living in nursing homes or other institutions and all functioned independently in daily-life activities. None had a history of neurologic or psychiatric disorder, learning disability, or other major medical problems (cancer, significant cardiovascular disease, kidney disease, and so forth). None had a history of alcohol or other substance abuse during the five years preceding their participation.

Procedures

Subjects were tested individually in a quiet room by one of five certified speech-language pathologists (the five coauthors of this report). Each examiner tested 12 of the 60 subjects. The administration procedures used by the examiners are presented in Appendix A. Subjects' responses were transcribed, coded, and scored on-line by the examiners, who coded and scored responses and delivered prompts according to the procedures described in Appendix B. Administrations of the test also were recorded on audiotape. When necessary, examiners listened to the tapes to resolve any questions they may have had during test administration about patients' responses or how some responses should have been coded or scored. During the test subjects were given no feedback regarding responses except for that dictated by the procedures. However, general encouragement (e.g. 'you're doing fine') was occasionally given.

(2) Assessment of the reliability of administration and coding procedures

Transcription reliability

To evaluate the reliability with which examiners orthographically transcribed subjects' responses to test items, one of the examiners listened to the audiotapes of two randomly selected tests per examiner (10 tests) and independently transcribed each response. Items for which the examiner and the reliability judge agreed that the subject had given only the designated Boston Naming Test name were not included in reliability calculations, because high levels of agreement on these responses would have inflated the reliability values obtained. Transcriptions that differed by one phoneme or more were considered disagreements. Then point-to-point percentage reliability for the transcribed responses was calculated for each test, using the formula,

$$\left(\frac{\text{Number of agreements}}{\text{Number of agreements} + \text{disagreements}}\right) \times 100$$

Overall reliability of examiners' response coding

To evaluate the reliability with which examiners coded subjects' responses to test items, one of the examiners served as the primary reliability judge. She independently coded each of the 60 tests, using the original examiner's transcriptions of responses, but without access to how the original examiner had coded the responses. Another examiner served as the secondary reliability judge. He independently coded 10 randomly selected tests, using the same procedures used by the primary reliability judge. As before, items for which subjects gave only the designated Boston Naming Test name were coded by the reliability judges but were not included in reliability calculations. Point-to-point reliability for each examiner was calculated using the formula given above.

Category by category reliability of examiners' response coding

Point-to-point reliability for each coding category was calculated between the original examiner's coding and the primary reliability judge's subsequent coding of responses of all 60 tests.

(3) Assessment of non-brain-damaged subjects' performance

Subject's initial responses and their responses to prompts for each test item were analysed to determine: (a) the total error rate for each test item; (b) the frequency of various categories of responses for the group; (c) the frequency of consistently-given other names for test items; and (d) the effects of prompts on subjects' ability to name test items. These data were accumulated by two of the investigators for the 60 non-brain-damaged subjects.

Results

Reliability

Per cent reliability between the original examiner's transcriptions of responses and those subsequently transcribed by the primary reliability judge ranged from 87% to 100% across the five examiners ($M = 93{\cdot}4\%$). These reliability levels were considered sufficiently high to permit us to calculate response coding reliability on the examiners' transcriptions of responses.

Interjudge reliability between the four other examiners and the primary reliability judge for response coding ranged from 86% to 95% ($M = 89{\cdot}1\%$; table 1). Interjudge

Table 1. Intrajudge and interjudge reliability for coding responses to Boston Naming Test items.

Examiner	Agreements	Disagreements	% Reliability
1	164	4	98
2	119	6	95
3	94	12	89
4	85	14	86
5	103	16	87

Reliability for Examiner 1 is intrajudge reliability. Those for the other four examiners are interjudge reliability.

reliability between the primary and secondary reliability judges was 93%. Intrajudge reliability (between the primary reliability judge's original scoring of tests administered by her and subsequently rescored by her) was 98%.

Category by category reliability ranged from 32% to 100% (table 2). The reliability for Unrelated Name and Don't Know categories was poor even though they were frequently occurring categories. The majority of disagreements were between Unrelated Name and Related Name and between Don't Know and Off Task categories. Wrong Part, Visual Misperception, Mispronunciation, and No Response categories occurred too infrequently to permit any strong conclusions about their reliability. The overall adjusted reliability of the coding system using a matrix agreement system that adjusts for chance agreements (Zwick 1988) was 91%.

Assessment of non-brain-damaged subjects' performance

The mean number of items correct for our 60 adults was 54·50, with a standard deviation of 3·52 and a range of 40–59. The correlation between our subjects' BNT total score and age ($r = 0{\cdot}10$) was not significant, but the correlation between total score and education was significant ($r = 0{\cdot}30$, $P < 0{\cdot}01$), but weak, accounting for only 9% of the variance.

Ninety-one per cent of subjects' responses were correct names. Only 20 of the 60 test items were named correctly by all 60 subjects (table 3). These were scattered throughout the first 40 items. Another 20 were named incorrectly by less than 10% of our subjects. These were scattered throughout the first 50 test items. The remaining 20 test items were named incorrectly by 10% or more of our subjects. These items are marked with asterisks in table 3. Ten were in the first 50 items and 10 were the final 10 items on the test. The increase in errors across the test was not regular, but the largest number of high error items was concentrated at the end of the test.

Half of all incorrect responses were related names (table 4). Related Name and Don't Know responses accounted for nearly 80% of incorrect responses. Of 3 600 responses given by our subjects only four were initially coded as Visual Misperception and three were coded as Wrong Part.

Ten items were called a consistent other name by 10% or more of our subjects (table 5). Only one of these names was a true synonym (mouth organ for *harmonica*).

Table 2. Point-to-point reliability for coding categories.

Category	Agreements	Disagreements	% Reliability
Correct name	301	5	98
Multiple attempt	17	4	81
Related name	150	8	95
Unrelated name	7	15	32
Off task	23	4	85
Visual misperception	3	1	75
Wrong part	3	1	75
Mispronunciation	5	0	100
Don't know	49	15	77
No response	2	0	100

Items for which the examiner and the reliability judge agreed that the subject had given only the designated BNT name were not included in reliability calculations.

Table 3. Number of incorrect responses to Boston Naming Test items.

Item	Number incorrect	Item	Number incorrect
1. bed	0	31. rhinoceros	10*
2. tree	0	32. acorn	4
3. pencil	0	33. igloo	1
4. house	6*	34. stilts	0
5. whistle	0	35. dominoes	4
6. scissors	1	36. cactus	0
7. comb	0	37. escalator	0
8. flower	1	38. harp	0
9. saw	0	39. hammock	1
10. toothbrush	1	40. knocker	6*
11. helicopter	2	41. pelican	6*
12. broom	0	42. stethoscope	2
13. octopus	0	43. pyramid	3
14. mushroom	4	44. muzzle	7*
15. hanger	0	45. unicorn	7*
16. wheelchair	0	46. funnel	3
17. camel	1	47. accordian	5
18. mask	0	48. noose	8*
19. pretzel	0	49. asparagus	4
20. bench	0	50. compass	32*
21. racquet	0	51. latch	26*
22. snail	0	52. tripod	11*
23. volcano	3	53. scroll	6*
24. seahorse	5	54. tongs	11*
25. dart	1	55. sphynx	12*
26. canoe	1	56. yoke	16*
27. globe	5	57. trellis	7*
28. wreath	0	58. palette	18*
29. beaver	11*	59. protractor	39*
30. harmonica	8*	60. abacus	27*

Table 4. Types of incorrect responses on the Boston Naming Test.

Type	Number incorrect	Per cent incorrect
Total incorrect	326	9
Related name	164	50
Unrelated name	16	5
Multiple attempts	10	3
Off task	29	9
Visual misperception	3	1
Wrong part	3	1
Mispronunciation	5	2
Don't know	94	29
No response	2	1

Three were less precise names than the name specified in the Boston Naming Test and six were outright errors (table 5).

Prompting subjects to give an additional response following an Off Task or Don't Know response was effective in eliciting correct names 5% and 8% of the time,

Table 5. Frequently given other names for test items.

Type	Frequency
Synonym	
mouth organ (harmonica)	7
Less Descriptive	
rope (noose)	9
lock (latch)	20
harness (yoke)	10
Error	
muskrat (beaver)	6
hippopotamus (rhinoceros)	7
compass (protractor)	12
protractor (compass)	8
easel (tripod)	6
easel (palette)	7

BNT designated names are in parentheses.

Table 6. Correct responses following initial incorrect responses and prompt.

		Correct after prompt	
Response type	Number of prompts	Number	Percentage
Related name	135	53	39
Multiple attempts	21	13	62
Off task	21	1	5
Don't know	63	5	8
Visual misperception	4	1	25
Wrong part	3	1	33

respectively (table 6). Prompting for an additional response following Multiple Attempts and Related Name responses elicited correct names 62 and 39% of the time, respectively. There were too few Visual Misperception and Wrong Part responses to permit any conclusions about the effectiveness of prompts following these responses. Prompts were not given following two types of responses—Unrelated Name and Mispronunciation. (Not all incorrect responses that should have been followed by a prompt received one, and occasionally an incorrect prompt was given by examiners. These items were not included in the evaluation of the effectiveness of prompts.)

Discussion

These results suggested that examiners can reliably code most categories of responses to Boston Naming Test items using the administration and coding procedures described herein. The average interjudge reliability of the coding system was 89%, and the intrajudge reliability was 98%. The overall reliability of the coding system, adjusted for chance agreement, was 91%.

Overall, 91% of the responses given by our non-brain-damaged subjects were correct. Because the difficulty of test items supposedly increases as one progresses through the test, one would expect that almost all incorrect names should occur late in

the test. This was not the case. Of the 20 test items named incorrectly by 10% or more of our subjects, half were in the first 50 items. Few incorrect responses were Visual Misperception or Wrong Part responses. This suggests that few of the drawings were perceptually difficult or confusing. However, it seems likely that some Related Name responses were caused by visual misperception or confusion, or to subjects responding before they had looked carefully at the details of the drawings (such as saying schoolhouse or apartment building for *house*, tripod for *easel*, hippopotamus for *rhinoceros*). Examiners should be aware that such responses may be given commonly by adults who have no history of language or cognitive impairment. If this test were to be revised we feel that confusing items such as *compass* and *protractor* should be eliminated and other responses should be considered acceptable, such as mouth organ for *harmonica*, lock for *latch*. The latter response was given by a third of our subjects.

The performance of our non-brain-damaged subjects suggests that prompting for an additional response following Multiple Attempts and Related Names is likely to be profitable with aphasic subjects, in terms of eliciting the correct name for test items. Prompting following other types of incorrect responses was not very helpful to non-brain-damaged subjects. Whether such prompts would be helpful to aphasic subjects remains to be seen. The frequency of Visual Misperception and Wrong Part responses was too low to permit confident conlusions about the effects of prompts following those categories of responses.

Our normative data generally are consistent with those of Van Gorp *et al.* (1986). The mean number of items correct for the Van Gorp *et al.* study of 78 adults was 54·31 and the mean for our 60 adults was 54·50. Van Gorp *et al.* did not find a significant correlation between BNT total score and either age or education, although they noted greater variability in the naming performance of the older age groups. Even though they found no significant correlation between total score and age, Van Gorp *et al.* reported standard deviations and cut-off scores for 'unimpaired performance' by age group. Cut-off scores for unimpaired performance in the Van Gorp *et al.* study ranged from 38, for the oldest group, to 51, for the youngest group. Like Van Gorp *et al.*, we did not find strong relationships between BNT total score and age or education. Consequently, there seems little point in constructing cut-off scores based on age or education. For our 60 adults, the cut-off score for unimpaired performance was 48 (using Van Gorp *et al.*'s criterion of two standard deviations below the mean). Persons with a score of 48 or worse would fall below the 5th percentile of our sample.

Part II: Revisions of prompting and coding procedures

Method

Because Don't Know and Off Task responses could not be differentiated readily from one another, and because prompts following both of these types of responses were ineffective in eliciting correct responses, we combined them into one category, called Associated Responses, which does not receive a prompt. We also combined the Unrelated Name and Related Name categories into one category (Other Name) because of the difficulty we had in differentiating these categories from one another. We retained the Related Name category prompt ('Tell me another name for that') for this combined category because it had been effective in eliciting correct repsonses

nearly 42% of the time. These changes seemed likely to increase the reliability of the coding system and to make the system easier for clinicians to use.

Two examiners who had been involved in the original study listened to 10 of the original 60 audiotaped administrations of the BNT and coded subjects' responses that were given before any examiner prompts, using the revised procedures. Any disagreements were discussed and a consensus reached on the correct coding for all responses. This consensus coding was used as a master coding protocol for the 10 tapes.

Next, four graduate students in speech and language pathology were recruited to learn the revised procedures. These students had previously given the standard Boston Naming Test but had not seen or used our revised procedures. These students were given only the printed administration and response coding and scoring instructions found in Appendices A and B, and were given no additional oral or written instruction or assistance as they learned the procedures and coded and scored the 10 audiotaped administrations of the Boston Naming Test.

Results

Reliability between the master coding protocol and the students' coding of responses was in the range 87%–92% ($M = 90\%$). Category-by-category per cent reliability for the four categories that occurred eight or more times in the 696 responses on which reliability was calculated were: Correct Name, 92%; Multiple Attempts, 86%; Other Name, 86%; and Associated Response, 91%. As before, items for which subjects gave only the designated Boston Naming Test name were coded by the students but were not included in reliability calculations.

Discussion

The revised and expanded administration instructions and response coding, scoring, and prompting procedures reported herein make the Boston Naming Test a more reliable and informative measure of confrontation naming. These revised procedures do not, in our opinion, compromise the validity of the norms provided by Kaplan *et al.* (1983) and Van Gorp *et al.* (1986), because they are consistent with the procedures apparently employed when these normative subjects were tested. If one wishes to use any of the published norms (Kaplan *et al.* 1983, Van Gorp *et al.* 1986; or those reported herein), probably only the responses specified in the BNT scoring booklet should be accepted as correct. However, when the results of such tests are interpreted, the clinician should consider that some BNT test items consistently elicit names other than those specified as correct in the test manual, and that other items are vague or confusing to nonbrain-damaged adults. Therefore, when brain-damaged adults misname these items, one should be cautious in attributing the errors to the patient's brain damage.

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Appendix A

Instructions for administering the Boston Naming Test

Begin with Item 1 and continue through to Item 60 or until the subject makes six consecutive errors. Give the following instructions:

> 'Tell me the *name* of each of these pictures. Some pictures may have more than one name. Tell me the most common name for each one. If you are not sure, just give your best guess.'

Write down the patient's responses in as much detail as necessary. Code the response and (when appropriate) give prompts based on the final utterance in the patient's response (e.g. 'Let's see, that looks like a church. No. *It's a house*').

Provide only one prompt (when approprate) per patient response to a test item, except for responses which follow Visual Misperception plus Stimulus Prompt, and Wrong Part plus Prompt. These responses are treated as first responses and may receive an additional prompt (see coding procedures).

If the patient does not appear to understand the task or seems to forget the task, the general prompt 'Tell me the *name* of the picture' may be given. However, do not give this prompt if the patient appears to understand the task, but is either editorializing about the task or attempting to self-cue by talking about the item. In this case, no prompt is given and the response is coded as an associated response (AR) (see coding procedures.)

Allow the patient 20 seconds to respond to each picture. If a patient continues to respond beyond 20 seconds, say 'Let's go on', and do not give an additional prompt. Code the final response.

If a prompt fails to elicit the correct response, a phonemic cue (the initial phoneme(s) underlined on the score sheet) may be given at the examiner's discretion to obtain additional information about the patient's ability to retrieve the word. However, providing phonemic cues might provide information to the patient that could affect his or her performance on subsequent administrations of the test.

Allow the patient 5 seconds to respond following a prompt or phonemic cue. Responses following Visual Misperception plus Stimulus Prompt and Wrong Part plus Prompt are treated as first responses and patients are therefore allowed 20 seconds to respond to these prompts.

After completing the test, score each item '+' or '−' (see coding and scoring procedures.) You will need to modify column headings on the score sheets to allow for marking of response codes and +/− scores.

Appendix B

Response coding and scoring procedures for the Boston Naming Test

Code	Patient response	Example of response	Examiner response
CN (Correct Name)			
	Picture name as in BNT test booklet.	house	Accept and go on Score +
	Multiple-word or compound-word responses containing the BNT name with same meaning as BNT name.	large house	Accept and go on Score +
	EXCEPTION. If a multiple-word or compound-word response contains the BNT name, but it has a different meaning, prompt and code it as Other Name.	doghouse apartment house	See ON, below Score −
ON (Other Name)			
	A name (or multiple names) that does not contain the BNT name, and which does not seem to be a visual misperception.	building dwelling school a kind of factory	'Tell me *another* name for that' Score − even if the correct name is given after the prompt
MA (Multiple Attempts)			
	Multiple attempts in which the BNT name is given, but is NOT the last response given.	house, building	'Tell me the *best* name' Score + if the correct name is given after the prompt
	EXCEPTION 1. If the response contains multiple attempts, and the BNT name is the last one, code it as a correct name.	building, house	Accept and go on Score +
	EXCEPTION 2. If the response contains multiple attempts that *do not* include the BNT name, code the final response and prompt as appropriate.	building, school	
AR (Associated Response)			
	If the response consists of description of use, personal reactions to the item, telling what the item is not, or an indication that the patient doesn't know the name of the item, score it as an associated response.	People live in it. That's a big one. It's not a store. It's like an apartment. I can't think of it.	No prompt Score −

Code	Patient response	Example of response	Examiner response
VM (Visual Misperception)			
	Name that suggests that the patient has misperceived the test picture.	coffin (for tongs) umbrella (for mushroom)	Stimulus prompt next to test item in BNT booklet Treat the response to the stimulus prompt as a first response and prompt as appropriate Correct responses following a stimulus cue are scored +
WP (Wrong Part)			
	Name for wrong part of the picture.	doorknob (for knocker)	'No, this' and point to part to be named Correct responses following a prompt are scored +
MP (Mispronunciation)			
	Articulatory errors, literal paraphasias, unintelligible words, or neologisms that do not result in other real words. Singular form for words whose standard form is plural (e.g. scissor, tong).	crushroom abiscus (abacus) scissor	Accept and go on For later scoring, score as + if a naïve listener would recognize it as the BNT name
NR (No Response)			
	No response in 20 sec.	—	'Give it a try' Score − even if the correct name is given after the prompt

Speech Pathology Section, Minneapolis Veterans Administration Medical Center. November 1987; revised May 1988.